COMPUTER VISION

ASSIGNMENT 2

Apply, Compare and Justify Image Enhancement and Segmentation Techniques

Smart Document Scanner for the College Library

Assignment details	Value
Course outcome	CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images
Total marks	50
Student name	Thikkavarapu Sanhith
Register number	192311228

TASK 1 — Basic Grey-Level Transformations (4 marks)

Point processing uses s = T(r): an output pixel depends on its own input intensity. Here the visible goal is dark, distinguishable text on a light paper background, while preserving faint pencil marks where possible.

Methods compared

Technique	Operation	Strength	Main risk / decision
A. Negative + log	Negative: s = 255 − r. Log: s = 255 log(1 + 9r/255) / log(10).	Negative is useful for reverse-polarity display; log expands some dark values.	Negative reverses the expected print polarity; log compresses bright paper values. Keep for special inspection, not the default export.
B. Gamma	s = 255(r/255)^γ; tested γ = 1.4.	γ > 1 darkens mid-tones and expands text/paper separation in this test.	One γ does not fit very light and very dark pages equally; clipping and tonal compression are possible.
C. Piecewise stretch	Map r ≤ 90 to 0; 90–180 linearly to 0–255; r ≥ 180 to 255.	Strong separation for the observed grey band; very simple LUT implementation.	Clips pencil/background variation outside the control points. Preserve the original before applying it.

Measured synthetic results

Method	Mean ink grey	Mean paper grey	Absolute mean gap	Global σ
negative	142.073	38.84	103.233	42.057
log	164.808	237.014	72.206	32.559
gamma_1.4	88.646	202.769	114.122	45.751
piecewise	96.855	250.636	153.782	61.974

The measured mean gap was largest for the 90–180 piecewise stretch (153.782 grey levels), followed by gamma 1.4 (114.122), negative (103.233) and log (72.206). These numbers come from the first synthetic page only and are not accuracy scores. The stretch gives the highest visible separation but may erase light pencil, stains or faint marginal notes by mapping them to 0 or 255.

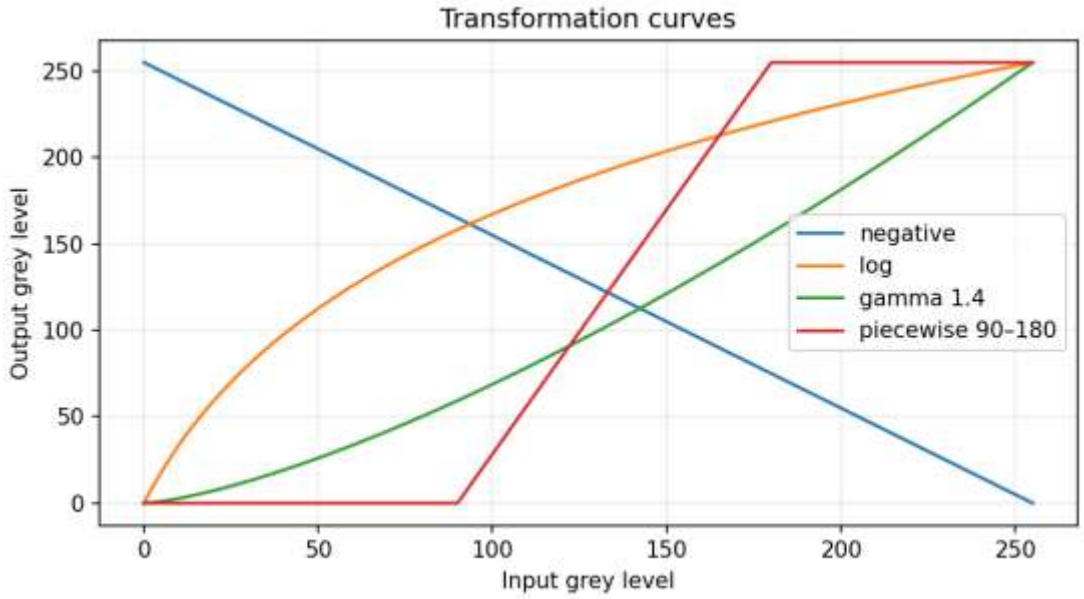


Figure 1. Mapping curves for the negative, log, gamma and piecewise transformations.

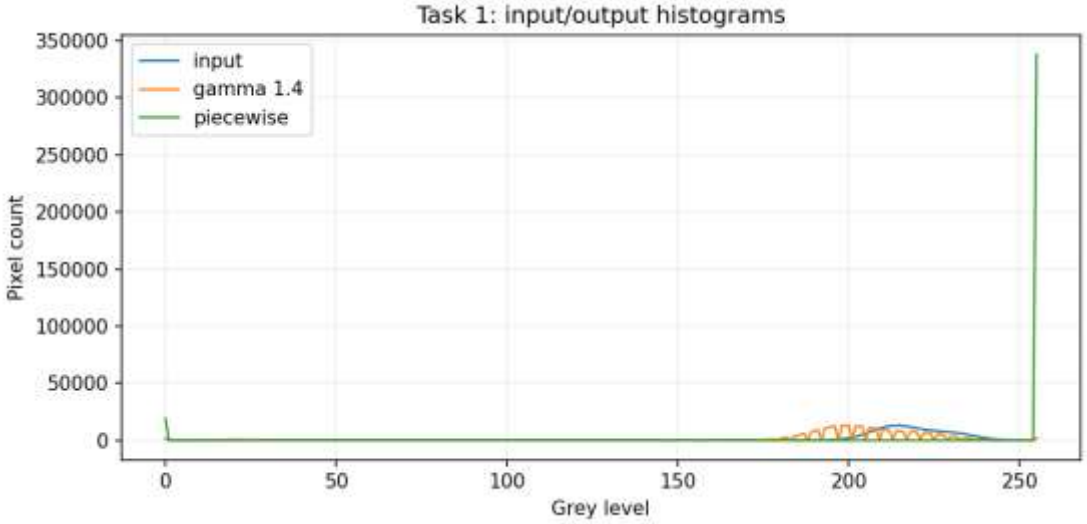


Figure 2. Input and enhanced-output histograms for the synthetic example.



Figure 3. Example outputs produced by the transformation candidates.

Bit-plane slicing and failure cases

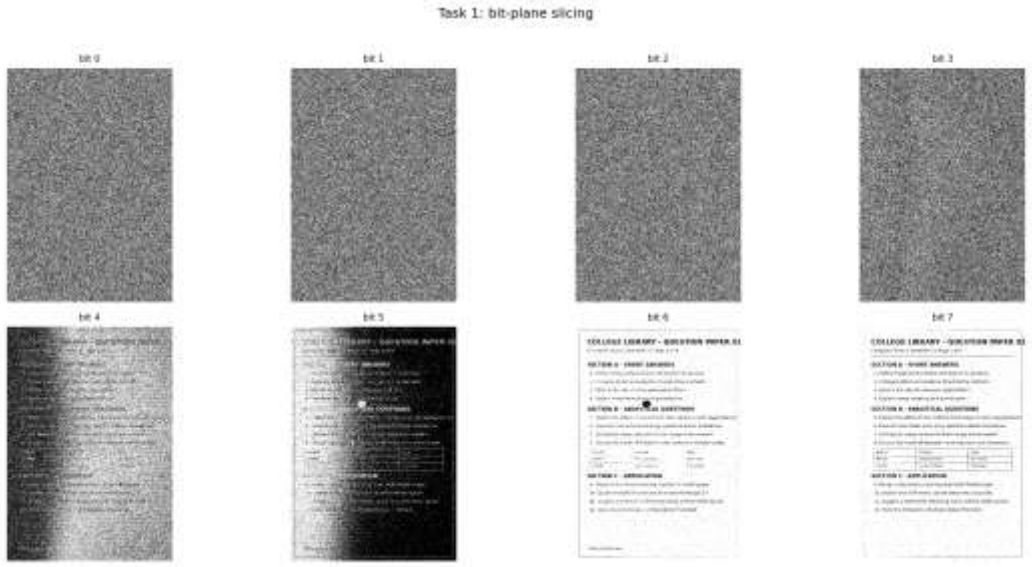


Figure 4. Bit 0–7 planes from one generated noisy page.

For typical 8-bit scans, low-order bit planes (0–3) are dominated by grain, JPEG/noise variations and small intensity fluctuations. Higher planes (roughly 5–7) preserve the broad paper/text structure, but also encode shadows and scanner bands. This is page-dependent; select planes by inspecting the actual bit-plane images, not by assuming a fixed set of planes is always best.

- A single γ on a very light page may darken text effectively but reduce paper texture contrast; on a very dark page it can turn both text and background dark and make local differences harder to separate. Estimate page tone first or use a per-page monotonic LUT.

- Piecewise stretching loses faint pencil marks if their intensity lies below the lower breakpoint or close to a clipped paper region. Use gentler control points or retain a separate colour/greyscale archival master.
- A pretrained enhancement model such as BEDSR-Net or DocEnTr may better handle combined shadow/blur, but it must be tested on the five same faded pages with OCR CER, faint-stroke preservation, latency and a check for altered strokes. Their checkpoints were not available in this run, so no model score is claimed [3,4].

Decision Use a conservative piecewise LUT for high-contrast print output only when the page histogram supports the 90–180 assumption. For archival OCR, start with CLAHE or gamma and test text/pencil retention. Always save the original and record LUT parameters.

TASK 2 — Histogram Processing (4 marks)

Histogram processing redistributes intensity values using the page-wide cumulative distribution or local tile statistics. It can improve visibility but cannot, by itself, know which variations are text, stains, texture or sensor noise.

Method	Observed synthetic evidence	Advantages	Limitations / selection
Global equalization	Mean global σ = 74.168; mean entropy = 5.799 bits across three representative pages.	Fast and can make weak global contrast more visible.	Changes blank paper strongly and amplifies noise; does not undo a spatial shadow. Use as a comparison baseline.
Histogram matching	Mean global σ = 39.379; mean entropy = 2.995 bits, using the first generated clean page as reference.	Can standardize appearance across a known acquisition protocol.	Highly dependent on a representative reference. This reference was dominated by paper levels and compressed useful variation; do not match to a mismatched sample.
CLAHE	clipLimit = 2.0, tileGridSize = 8×8; mean entropy = 6.747 bits and global σ = 43.693.	Local contrast recovery helps centre/edge lighting variation; clipping limits over-amplification.	High entropy can mean amplified noise, not only better text. Large clip limits create grain; small tiles can produce texture/tile artefacts.

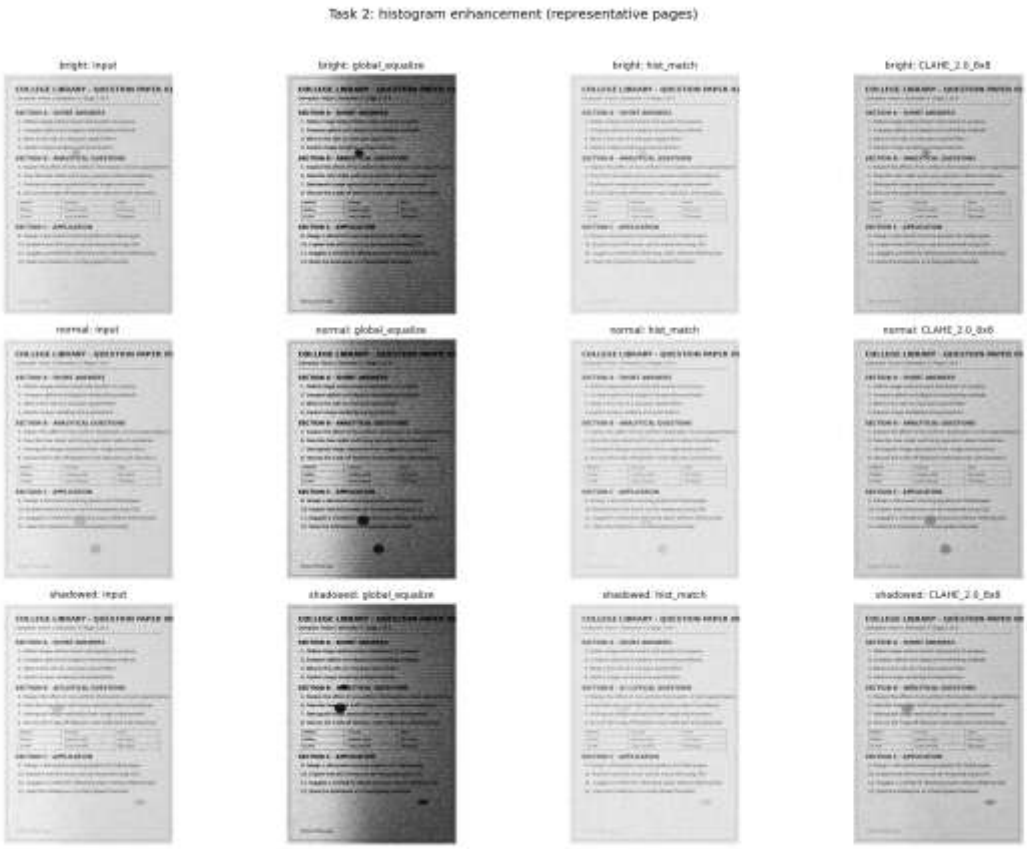


Figure 5. Representative histogram-based outputs on three synthetic page conditions.

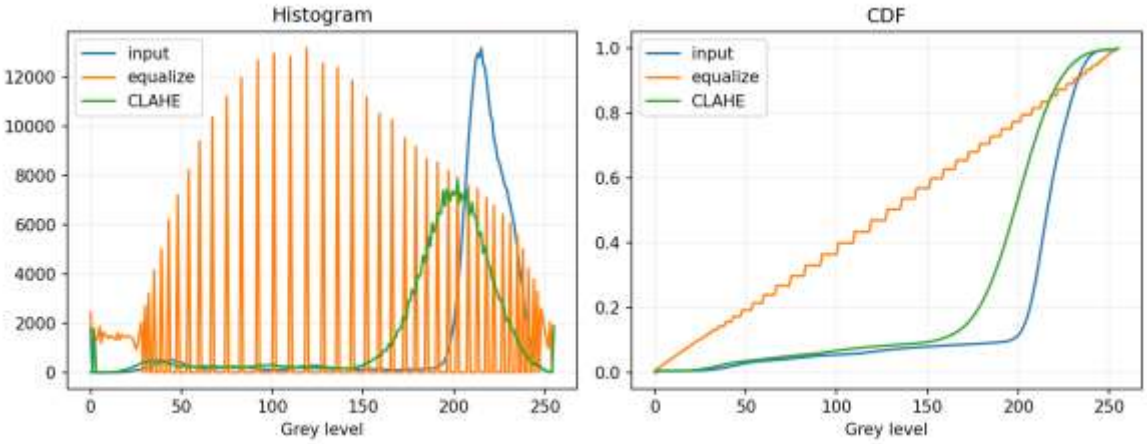


Figure 6. Input/output histograms and cumulative distribution functions.

Local mean/variance and parameter choice

Global equalization uses one mapping for every pixel. If the left side is shadowed and the centre is bright, one global mapping cannot make both local paper regions equally white without also stretching noise. The local mean $\mu(x,y)$ and variance $\sigma^2(x,y)$ from a 31–41 pixel neighbourhood should be plotted across the page: a systematic change in μ across x is evidence of illumination variation, whereas high variance in an otherwise blank region suggests grain, dust or texture.

- Start with CLAHE clipLimit = 2.0 and tileGridSize = (8,8) on a normalized working image; compare (4,4), (8,8) and (16,16) using Dice/OCR and noise in blank regions. A page-normalized grid should be adapted to the input resolution.
- For a page that is about 90% blank, the histogram is dominated by paper. Global equalization may use many output levels for background differences while text occupies few pixels; visual contrast may increase but faint letters or pencil may not improve.
- An excessively high CLAHE clip limit makes noise and the faint every-six-row scanner bands locally prominent. If entropy rises but Dice/CER does not improve, lower the clip limit or remove periodic bands separately.

Decision Use CLAHE with moderate clipping for variable mobile lighting, then measure blank-paper standard deviation as well as text contrast. Histogram matching is appropriate only when the reference page was captured and aged similarly to the target.

TASK 3 — Spatial Filtering and Image Smoothing (5 marks)

The test adds Gaussian noise ($\sigma = 8$ in the controlled noise-only experiment) and 1% salt-and-pepper noise to a clean generated page. PSNR and SSIM are calculated relative to that clean reference; runtime is for the 512 × 724 demo and is hardware-dependent.

3 × 3 measured comparison

Filter	PSNR (dB)	SSIM	Time (ms)	Interpretation
box3	20.3939	0.64818	0.168	Lowest SSIM here; averages pixels and can fade small dots/strokes.
gaussian3	22.2716	0.70431	0.34	Highest PSNR in this mixed-noise trial; smooth grain without strong shape changes.
median3	20.0016	0.70384	0.193	Strong for impulse noise; SSIM close to Gaussian while preserving edges better than an average.



Figure 7. Box, Gaussian and median smoothing outputs.

Kernel-size sweep

Filter	Kernel	PSNR (dB)	SSIM	Time (ms)
box	3x3	20.3939	0.64818	0.168
gaussian	3x3	22.2716	0.70431	0.34
median	3x3	20.0016	0.70384	0.193
box	5x5	18.619	0.61039	0.177
gaussian	5x5	20.7632	0.69535	0.266
median	5x5	17.7504	0.65945	0.664
box	7x7	18.0598	0.59437	0.254
gaussian	7x7	19.6426	0.67797	0.274
median	7x7	17.0404	0.64866	10.298

Task 3: kernel size sensitivity

noisy



box 5x5



gaussian 5x5



median 5x5



box 7x7



gaussian 7x7



median 7x7



Figure 8. Visual impact of 5×5 and 7×7 kernels.

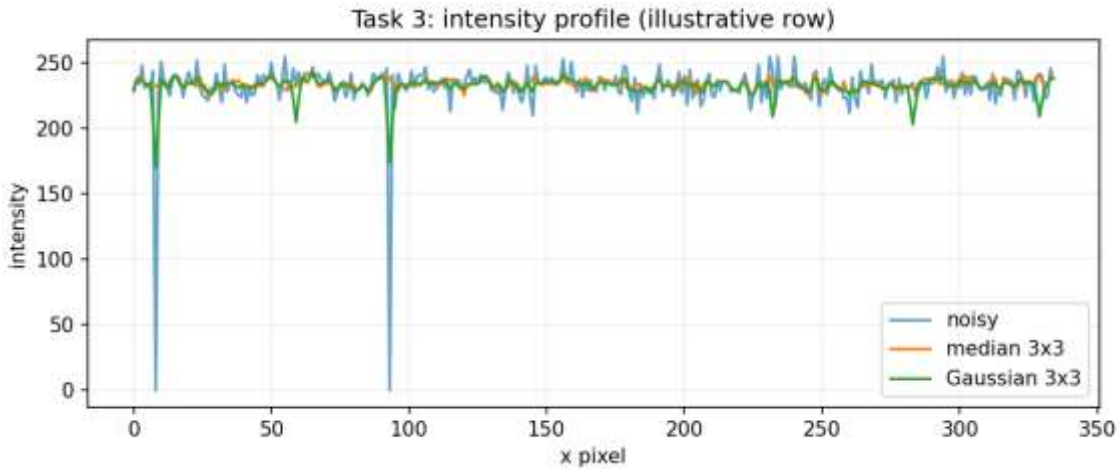


Figure 9. Row intensity profile: noisy input versus filtered outputs.

Correlation vs convolution: worked 3 × 3 example

Input patch P = [[10,20,30],[40,50,60],[70,80,90]]. Use the directional kernel K = [[1,2,1],[0,0,0],[-1,-2,-1]]. Correlation is Σ(P·K) = 80 − 320 = −240. Convolution rotates K by 180°, so the result is +240. OpenCV cv2.filter2D implements correlation (it does not flip the kernel). A small numerical example is also saved in manual_filter_example.txt inside the code package.

Use BORDER_REFLECT for filtering page edges: it mirrors nearby content and avoids creating a hard black padding edge. Zero padding may introduce an artificial dark frame and trigger false page-border/edge detections.

- A 7×7 box filter averages across thin 2–4 pixel strokes, lowering their peak contrast and sometimes merging adjacent characters. Increasing the kernel to remove dust is not a safe substitute for component-area filtering.
- Skipping smoothing leaves grain and the 6-row scanner band visible to Sobel/Canny and global thresholding; smoothing should be small enough not to delete punctuation.
- For this mixed synthetic noise, Gaussian 3×3 gave PSNR 22.272 dB and SSIM 0.704; median 3×3 gave SSIM 0.704 and is especially useful for the impulse component. A practical baseline is median 3×3, followed by a light Gaussian only if blank-region variance remains high.
- DnCNN is a suitable deep denoising comparator, but it was not executed because a compatible pretrained checkpoint and the real page dataset were not supplied. Compare PSNR/SSIM, OCR CER, time and small punctuation preservation before deployment.

Decision Begin with a 3×3 median for impulse noise; reject 7×7 as the default. Use measured OCR/SSIM and character connectivity to decide whether extra Gaussian smoothing is justified.

TASK 4 — Spatial Filtering and Image Sharpening (5 marks)

Sharpening should strengthen an already reasonably clean character edge. It cannot reconstruct text that was never captured, and it raises noise and halo risk when applied aggressively.

Technique	Equation / OpenCV route	Measured results	Assessment
Laplacian sharpening	$g = f - c\nabla^2 f$; cv2.Laplacian in float, then subtract with c = 0.45.	Variance of Laplacian = 3370.524; blank-region σ = 11.041; PSNR to clean = 24.823 dB.	Very strong high-frequency response; the sharpness number can be inflated by noise. Watch halos.
Unsharp / high-boost	$g = f + k(f - \text{blur}(f))$; Gaussian blur then cv2.addWeighted.	Variance of Laplacian = 558.588; blank σ = 10.093; PSNR = 22.863 dB.	More controlled general-purpose choice; start with k = 0.5–0.8.
Gradient sharpening	Add scaled Sobel magnitude $\lvert\nabla f\rvert$ to the page.	Variance of Laplacian = 907.838; blank σ = 7.103; PSNR = 18.972 dB.	Emphasizes edges but can make characters look outlined and may increase noise near every transition.



Figure 10. Laplacian, unsharp-mask and gradient-sharpened outputs.

Interpretation and parameter choice

A second derivative responds to changes in gradient on both sides of an edge; a thin stroke contains closely spaced transitions, so the Laplacian may emphasize it strongly. The same response appears around grain and stains. First derivatives (Sobel) estimate direction/magnitude and can be easier to gate by edge orientation, but their gradient magnitude also reacts to noise. A defensible sequence is a small denoising stage, then a mild sharpening stage, then thresholding.

- A large high-boost factor k on an unsmoothed image increases the high-frequency component and creates bright/dark halos, noise amplification and false dots. Reduce k or skip sharpening when the original is already crisp.
- A Laplacian computed as signed CV_32F can contain negative values; converting directly to uint8 clips negative responses and hides half the edge information. Normalize or offset only for display; use the signed result correctly for arithmetic.
- The table shows that the highest Laplacian variance does not automatically mean the best OCR. Check character error rate, punctuation preservation and blank-region noise, not a sharpness metric alone.
- A pretrained deblurring/super-resolution model should be tested on the same five blurred phone images against unsharp masking. Measure CER/WER and traceable edge preservation. No pretrained model result is invented here because model weights and matching real photos were not available.

Decision Use unsharp mask at low strength only where a sharpness/blur metric shows softness. For an archival scanner, avoid super-resolution outputs as a sole evidence image because generated strokes may alter the original text.

TASK 5 — Frequency-Domain Low-Pass Filtering (5 marks)

A DFT pipeline pads the image, computes and centres F(u,v), multiplies by H(u,v), inverse-transforms and crops. Ideal filters have a sharp cut-off; Butterworth filters soften the transition with order n; Gaussian filters have no sharp cut-off and usually show less ringing.

Filter	D_0	Global σ	Entropy (bits)	Laplacian variance	Time (ms)
ideal	15	14.17	5.7079	1.102	388.814
ideal	30	15.746	5.8809	1.421	161.139
ideal	60	21.759	6.3393	2.317	125.93
butterworth	15	13.72	5.5685	0.847	219.489
butterworth	30	15.298	5.7676	1.197	193.094
butterworth	60	19.187	6.0066	2.106	202.745
gaussian	15	13.803	5.5561	0.84	188.84
gaussian	30	15.613	5.7854	1.261	153.936

Filter	D_0	Global σ	Entropy (bits)	Laplacian variance	Time (ms)
gaussian	60	19.658	5.9973	2.922	178.536

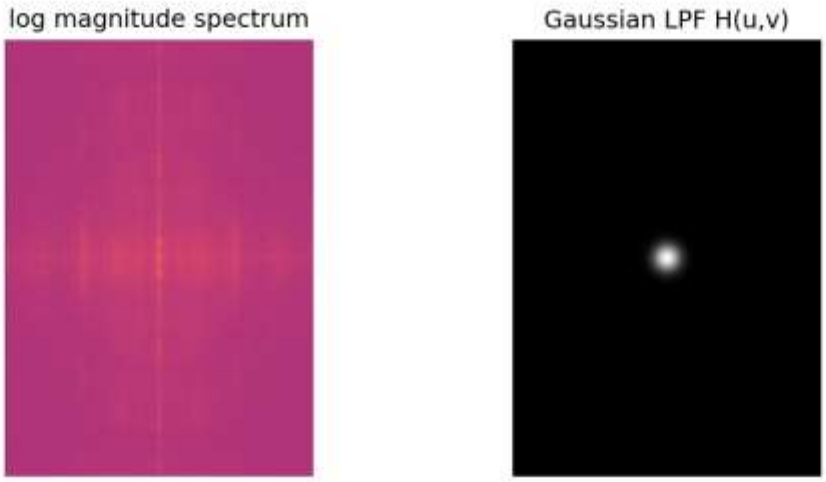


Figure 11. Log-magnitude spectrum and Gaussian low-pass transfer function.



Figure 12. At $D_0 = 30$, all three low-pass outputs blur thin text strokes.

OCR is the deciding measurement

Image/filter	Character error rate (CER)	Tesseract time (ms)
input	0.19368	518.986
ideal_D0_30	0.99789	103.928
ideal_D0_60	0.90632	340.784
butterworth_D0_30	0.99895	98.467
gaussian_D0_30	0.99895	105.951
gaussian_D0_60	0.90737	353.96

On the synthetic reference transcript, the unfiltered input CER was 0.19368 (about 19.4%). Applying ideal, Butterworth or Gaussian low-pass filtering at $D_0 = 30$ raised CER to approximately 0.998–0.999; $D_0 = 60$ still raised CER to about 0.906 for ideal/Gaussian. This is a useful result against a common

assumption: suppressing high-frequency stripes with a broad low-pass filter also removes high-frequency text strokes. In this test, the filter makes OCR far worse.

- Ideal LPF can create ringing near sharp text edges due to its discontinuous frequency response. Butterworth reduces but does not eliminate ringing; Gaussian is generally the smoothest, but all can blur characters when D_0 is too small.
- Without padding, the DFT treats opposite edges as adjacent and causes wrap-around seams. Reflect padding reduces this discontinuity; tapering or additional padding may be needed for large border artefacts.
- A frequency-domain Gaussian and spatial-domain Gaussian are mathematically related by the Fourier transform, but discrete kernels, border handling and crop/pad policies produce small differences. Compare MAE/PSNR between actual outputs on the same image rather than assuming bitwise equality.
- For scanner bands every six rows, locate narrow spectral peaks at the corresponding periodic frequency and use a notch/band-reject filter, or estimate and subtract row-wise brightness variation. Retain a broad low-pass only if OCR evidence confirms no text loss.

Decision Do not retain broad low-pass filtering in the OCR path based on the current evidence. Use targeted notch or row-profile correction and re-run Tesseract CER plus visual edge checks against ground truth.

TASK 6 — Frequency-Domain High-Pass Filtering (4 marks)

High frequencies describe stroke transitions, while low frequencies contain slow changes such as illumination shadows. A plain high-pass image discards the DC/overall paper tone, so it is rarely suitable as the final scan. High-frequency emphasis or homomorphic filtering is more useful when the goal is to reduce illumination variation while retaining readable background brightness.

Method	Mean output grey	Global σ	Blank-region σ	Time (ms)	Interpretation
ideal	161.516	25.092	15.082	21.676	High-pass output is display-scaled; paper brightness is not meaningful.
butterworth	162.842	24.715	14.983	33.615	Smoother transition than ideal, but direct high-pass is still not a natural page image.
emphasis	144.598	48.846	28.919	34.419	Retains part of the low-

Method	Mean output grey	Global σ	Blank-region σ	Time (ms)	Interpretation
					frequency tone and boosts edges; raises blank-region noise.
homomorphic	162.527	43.771	25.957	41.07	Best candidate for uneven illumination, but the measured blank-region σ shows noise can be amplified.



Figure 13. High-pass filters versus homomorphic enhancement.

Photometric model and failure modes

Using $f(x,y)=i(x,y)r(x,y)$, the logarithm gives $\log f = \log i + \log r$. The illumination term is generally smooth/low frequency; reflectance has more edges. A homomorphic filter uses $\gamma L < 1$ and $\gamma H > 1$ (tested 0.7 and 1.5) to compress slow illumination variation and enhance edges. If γL and γH are swapped, the illumination component gets boosted while the edges are suppressed, producing a less even, softer page.

- Applying an unmodified ideal HPF output as the scanned page removes paper tone and converts the page into an edge map. It should only be used as an intermediate representation.
- High-frequency emphasis preserves some brightness but also boosts grain, every-six-row stripes and JPEG boundaries. Use a small gain and denoise first.
- A document-specific learned shadow removal model such as BEDSR-Net is a reasonable comparator; the published CVPR paper reports a model built specifically for single-document shadow removal and provides paired synthetic shadow data [3]. The model was not benchmarked locally because weights/data were unavailable; evaluate shadow uniformity, text contrast, noise, CER and time on the same shadowed photos.

Decision Prefer flat-field/background estimation or restrained homomorphic correction before binarization. Never substitute a high-pass edge image for the archival page.

TASK 7 — Point, Line and Edge Detection (5 marks)

The detector should match the object type: isolated dust is point-like; table rules are long and directional; the page boundary is a large closed quadrilateral. One edge map should not be interpreted as all three feature classes.

Detector	Precision (±2 px)	Recall (±2 px)	Observed behaviour
Laplacian_point_mask	0.0543	0.2277	Only 5.4% precision; many text/noise pixels also trigger the point response. Component area and shape filters are needed.
morphological_lines	1.0	0.9843	1.000 precision and 0.984 recall on the generated horizontal/vertical line mask; the synthetic page has axis-aligned ruled lines.
Sobel_gradient	0.1259	1.0	Full recall but only 0.126 precision because letter edges are mixed with line/border pixels.
Canny	0.1749	1.0	Full recall but 0.175 precision against the

Detector	Precision (±2 px)	Recall (±2 px)	Observed behaviour
			line mask; Canny detects many character contours as well as rules.

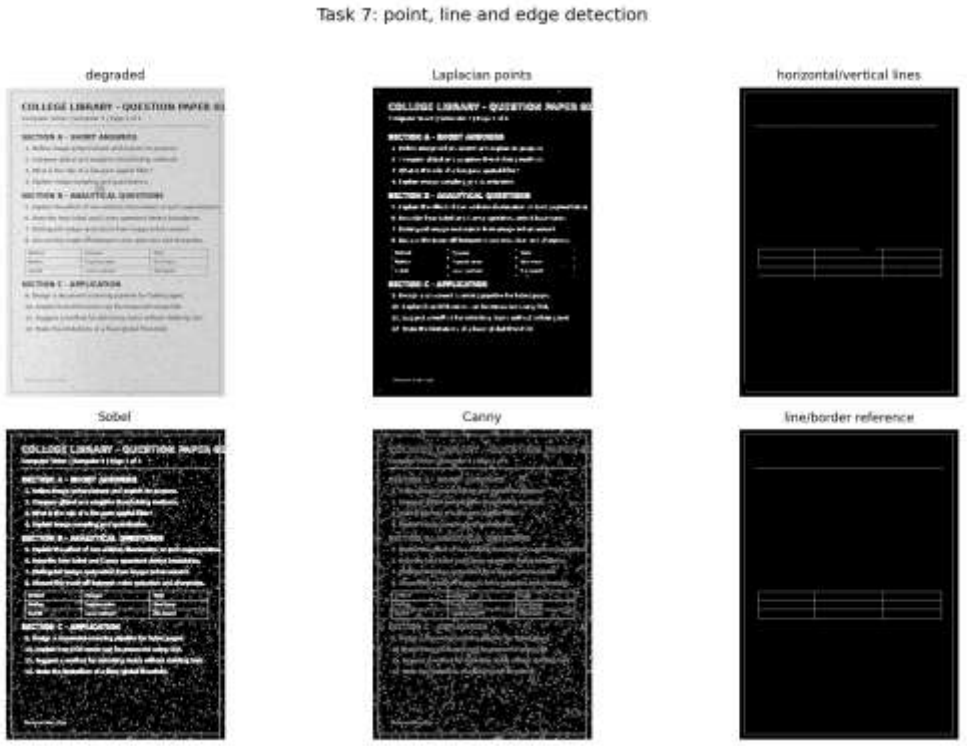


Figure 14. Point mask, directional line morphology, Sobel and Canny outputs.

Thresholds, tilt and clutter

- A Laplacian point threshold of $|R| \geq 95$ was used in the synthetic run. The low precision shows that one fixed T is not enough. Apply 3×3 median smoothing, connected-component area/compactness filters, and a local paper-baseline residual before declaring dust.
- Directional morphology used 35×1 and 1×35 structuring elements after thresholding. It was accurate on this axis-aligned synthetic rule set. A 2–5 degree page tilt breaks horizontal/vertical continuity; deskew first or use a rotated structuring element / Hough transform over a range of angles.
- Do not treat every character as a point; letters are naturally made of multiple small dark components. Search for isolated compact components in blank margins, exclude text lines, and require a plausible area/radius.
- Use Canny after ROI restriction for page borders, then fit four sides with contours/Hough lines; use morphology for table rules. A pretrained HED detector could be compared on identical hand-annotated masks, but no HED checkpoint was run in this synthetic benchmark.

Decision Use directional morphology for ruled/table lines; use Canny plus largest-quad/geometric checks for page borders; use a separate blob detector for dust. Tune T and Canny hysteresis thresholds on hand-labelled real pages.

TASK 8 — Thresholding and Binarization (4 marks)

The text mask should be the foreground (white in the evaluation mask) and paper the background. A wrong polarity must be corrected before computing Dice. Dice = 2TP / (2TP + FP + FN).

Method	Mean Dice	Range	Sample count
iterative_global	0.6945	0.6897–0.6964	10 pages
Otsu	0.6927	0.6875–0.6955	10 pages
adaptive_gaussian	0.6843	0.6801–0.6873	10 pages
smoothed_Otsu	0.8294	0.8154–0.8425	10 pages
CLAHE_smoothed_Otsu	0.8311	0.8228–0.8431	10 pages



Figure 15. Thresholding outputs and a generated foreground reference mask.

Interpretation and parameter tuning

In the synthetic set, smoothed Otsu averaged Dice 0.8294; CLAHE followed by Gaussian 3×3 smoothing and Otsu averaged 0.8311 (range 0.8228–0.8431). Plain Otsu and iterative global thresholding averaged about 0.693–0.695, and adaptive Gaussian thresholding with block size 35 and C = 9 averaged 0.684. None reached the required Dice ≥ 0.90, so this benchmark does not support a claim of target compliance. Anti-aliased glyph masks, stain overlays, random noise and shadow also mean parameter tuning and mask conventions matter.

- A global threshold assumes one useful separation between text and paper. Under $f = i \cdot r$, spatially varying illumination changes paper values even when paper reflectance r is uniform; a single T can delete shadowed text or mark a dark shadow as foreground.
- Otsu chooses T by maximizing between-class variance and is fast when text/paper form separable intensity classes. Mild Gaussian smoothing improved Dice strongly in the synthetic set by suppressing grain before Otsu.
- Adaptive thresholding estimates a local threshold, but its block size must cover the background neighborhood while not treating individual characters as their own illumination region. Too small a block (e.g., smaller than a character-height neighborhood) fragments strokes and creates false foreground patches.
- On a real page with strong shadows, choose CLAHE + lightly smoothed Otsu as the tested baseline and switch to adaptive binarization if the shadow/brightness test set demonstrates higher Dice/CER. Ground-truth performance, not preference, determines the final method.
- A deep document-binarization model trained/evaluated on DIBCO must be compared on the same page masks with Dice, text preservation, stain artefacts, connectivity and latency. DIB Platform provides access to photographed and real-world document benchmarks and quality/time assessment [1,2]. No deep checkpoint was executed here.

Decision Use CLAHE + Gaussian 3×3 + Otsu as the first classical candidate on this dataset, but treat Dice 0.831 as below specification. Tune against real ground truth and use adaptive thresholding only where it demonstrably improves shadowed-page results.

TASK 9 — Region-Based Segmentation, Edge Linking and Integration (6 marks)

Page-boundary extraction and stain segmentation are different problems. The first should find a closed, approximately quadrilateral border even in shadow; the second should mark uncertain regions for review without confusing printed characters, paper texture and stains.

Candidate	Synthetic stain IoU	Dice	Result
region_growing_seeded	0.14967	0.26037	synthetic benchmark; seed-based method requires user seed
split_merge_proxy	0.08723	0.16046	synthetic benchmark
edge_contour_regions	0.02575	0.05022	synthetic benchmark



Figure 16. Region-growing, split/merge proxy, contour regions and stain reference.

Comparison and failure analysis

- Region growing starts at a seed and adds adjacent pixels within a grey-level tolerance. The seeded demonstration reached stain IoU 0.1497; that is not high enough for automatic deletion. A seed on a character grows across text; a seed on a stain may leak into similarly toned paper. Add gradient barriers, size limits and manual correction.
- Split/merge methods divide high-variance regions and merge adjacent uniform regions. The simple local-residual/connected-component proxy reached IoU 0.0872, showing that coarse thresholds do not resolve stain and text ambiguity. A true quadtree implementation needs a specified uniformity predicate, minimum cell size and merge-distance tolerance.
- The contour-region candidate reached IoU 0.0257 because thresholding groups many dark text/line components with possible stains. This is useful evidence that foreground masks are not stain masks. Reject it for stain classification.
- For page corners, threshold the likely paper region, close small gaps, select the largest external contour, approximate a convex quadrilateral, order the corners and use cv2.getPerspectiveTransform. If the border is interrupted, connect line fragments with morphology/Hough and check the quadrilateral area/aspect ratio.
- Measure corner error as mean Euclidean distance between each detected and annotated corner; measure stain IoU = intersection/union. A promptable model such as SAM 2 or watershed may help separate uncertain regions, but prompt count, IoU, latency and manual correction must be measured on the same images. SAM 2 was not run because weights and image annotations were not supplied.

Integrated pipeline diagram

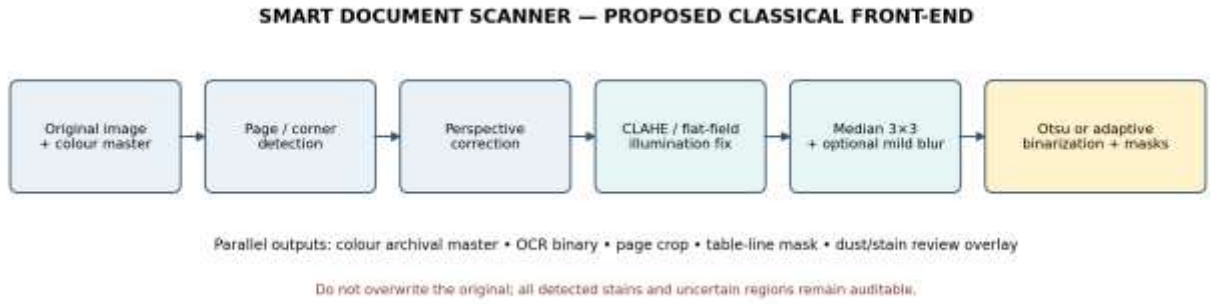


Figure 17. Proposed processing architecture; archival colour/greyscale master remains separate.

Design decision table

Decision	Evidence-based justification	What if not selected?
Keep original master	Needed for traceability and to verify text that an enhancement might change.	A binarizer or neural model could destroy faint marks permanently.
Page crop / perspective correction	A closed four-corner contour supports document coordinates and stable downstream processing.	Page skew makes rules slanted and changes OCR character geometry.
CLAHE (clip 2, 8×8) or flat-field	Local brightness changes; CLAHE entropy and visual contrast increased in the synthetic benchmark.	Global thresholding may fail under side shadow; excessive CLAHE amplifies blank-area noise.
Median 3×3	Small impulse specks exist and small kernel preserves more character detail than 7×7.	Noise becomes false edges/dots, or larger blur deletes punctuation.
Gaussian 3×3 only when needed	A controlled mixed-noise test yielded PSNR 22.272 dB and SSIM 0.704 at 3×3.	Over-smoothing makes 2–4 px strokes thinner or joined.
Otsu after mild smoothing, with adaptive fallback	Mean synthetic Dice 0.831 for CLAHE + smoothing + Otsu; still below the 0.90 target.	Unvalidated method may fail on severe shadow; keep a target-compliance gate.
Morphological rules + Canny/quad	Line morphology precision 1.00 and recall 0.984 for generated axis-aligned rules; Canny has high recall but text false positives.	One generic edge detector mixes table lines, characters, dust and border.
Stain review overlay, not auto-delete	Synthetic stain IoU of even the best tested simple candidate was only 0.150.	Automatic cleaning could remove real ink or leave hidden stains unlabelled.
Stage timer + parameter log	Pipeline timing must be recorded on target PC; save settings and	The 3 s/page limit and reproducibility cannot be

Decision	Evidence-based justification	What if not selected?
	output hashes for audit.	confirmed.

Stage timing sample

Stage (512×724 synthetic input)	Measured time (ms)
read/decode (excluded file I/O)	0.0
CLAHE	0.624
median denoise 3x3	0.089
adaptive threshold	2.533
Canny edges	1.02
contours / annotations	28.058
total_core_pipeline	32.324

The measured core pipeline took about 32.3 ms on this small sample in the recorded run, dominated by contour processing. This does not prove 3 seconds/page at 2480 × 3508 because operations, memory bandwidth, I/O and target hardware differ. Run at full acquisition resolution and report median/p95 over at least 30 pages.

Decision Use edge linking/quadrilateral fitting for page borders and directional morphology for table rules. Treat stain detection as advisory until real annotated pages demonstrate acceptable IoU and false-negative rates.

TASK 10 — Adapt the Scanner to Phone Photographs (8 marks)

The new requirement changes the operating domain: 4000 × 3000 colour JPEGs (12 MP), angled page geometry, classroom-light shadows, blue handwritten notes that must survive, JPEG block artefacts, and a 1-second limit on the same CPU-only lab PC. The original assumption of a flat 8-bit grayscale A4 scan is no longer valid.

Revised constraints and design implications

Changed assumption	Implication	Response
Resolution: 512/2480×3508 → 4000×3000	12 MP has 12 million samples; full-resolution filtering and memory traffic can exceed 1 s on some CPUs.	Detect page at reduced working size; transform the original only after corner detection; preserve high-res archive master.
Grayscale-only input → colour JPEG	Grayscale may lower/merge blue pen and loses chromatic evidence.	Keep colour original; use RGB red channel (OpenCV BGR index 2) or channel differences to detect blue

Changed assumption	Implication	Response
		ink; create a separate OCR binary.
Flat scan → angle/perspective	Character heights and table rules vary across the frame.	Find four page corners and use perspective rectification.
Scanner grain/stripes → JPEG 8×8 patterns and tube lighting	Compression blocks, colour noise and shadow edges can create false points/lines.	Avoid broad low-pass; use mild denoise, local illumination correction and JPEG-aware testing.
3 s/page → 1 s/page	More aggressive time budget.	Time all stages; reduce working resolution for detection; process only the warped page ROI. Do not add homomorphic FFT unless its measured benefit offsets latency.

Measured comparison on generated phone photos

Strategy	Mean text Dice	Blue-note recall	Mean page-corner error	Mean time (ms)
A. Retain grayscale + Otsu	0.1026	0.0000	Not estimated	0.717
B. Modify: detect/warp, red channel, CLAHE + adaptive	0.6741	0.7639	0.614 px	6.049
C. Redesign: warp + homomorphic + adaptive	0.5753	0.7639	0.614 px	44.074

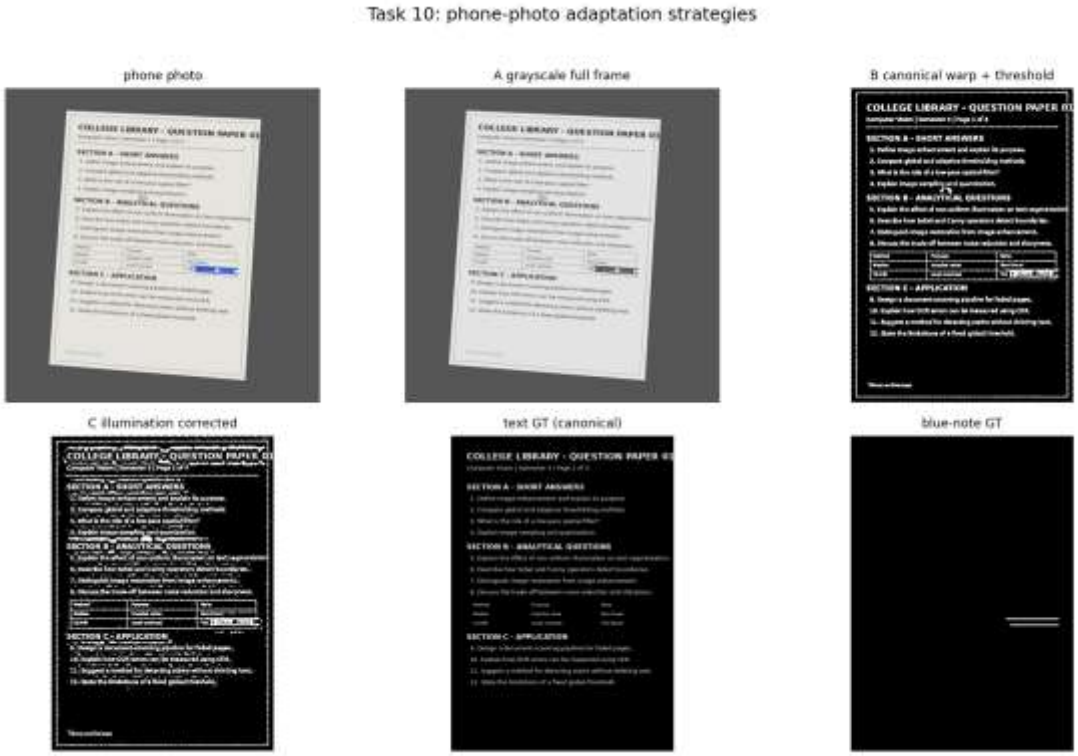


Figure 18. Perspective/JPEG phone-photo simulation and strategy outputs.

In the six-page synthetic phone trial, Strategy B had the best mean text Dice (0.6741), blue-note recall (0.7639) and mean corner error around 0.614 pixels, while requiring about 6.05 ms at the reduced 512×724 rectified output. Strategy C increased runtime to about 44.07 ms and did not improve Dice or blue-note recall; this argues against homomorphic FFT as a default on the CPU. Strategy A was fast but lost page geometry and blue-colour evidence, and its text Dice was only 0.1026 because the full dark background was binarized as foreground.

The times are for a 900×900 generated JPEG canvas and a 512×724 processing image, not for 4000×3000 camera frames. They cannot be extrapolated as a certified one-second result. Benchmark the real input size on the lab PC, including decoding, page detection, warp, output writing and OCR.

Recommended RETAIN / MODIFY / REDESIGN choice

Choice	Recommendation
RETAIN	Keep the original image as an archival master, use OpenCV/NumPy, store per-stage timings and masks, and retain the classical contour/warp logic.
MODIFY — chosen	Downsample a copy for corner detection; detect and order page corners; warp the original or a suitably scaled copy; keep RGB; use red-channel/blue-difference logic for blue notes; apply CLAHE with restrained settings and adaptive/global threshold selected by validation; export the colour page and a separate OCR mask.
REDESIGN only if validated	Use frequency-notch correction for visible JPEG/scanner periodicity or a fast illumination-background estimator if

Choice	Recommendation
	local tests show benefit. Avoid unconditional homomorphic filtering since the measured CPU time and Dice were worse in this simulation.

Document AI role, reliability and stopping condition

- LayoutLMv3 or Donut can assist downstream layout interpretation (titles, question numbering, table cells and reading order) after geometry and OCR are stable. They do not replace the need to preserve the colour source, rectify page geometry or generate an auditable binary/region mask for the stated requirements.
- Few-shot/foundation-model adaptation may reduce annotation demand but requires representative examples across cameras, lighting, paper colour and ink types. Validate on held-out pages and version the model/checkpoint, preprocessing, prompts, thresholds and evaluation masks.
- False acceptance means deleting/losing a faint handwritten note; false rejection means unnecessary manual review. Preserve both the original and enhanced output, add overlays for uncertain pixels, and prohibit automatic removal of handwriting or stains without a reversible review step.
- The recommended approach stops being suitable if the page border is completely occluded or a blue note is nearly the same intensity/chromaticity as the paper, or if the actual 12 MP end-to-end test exceeds 1 second. In those cases use user-assisted corner selection, retain the colour master for recovery and reallocate computation to the page ROI or better acquisition.

Decision MODIFY, rather than retain unchanged or immediately redesign. Strategy B gave the best synthetic trade-off; the one-second requirement remains unverified at 12 MP and must be demonstrated on the specified lab PC before release.

Final Conclusion

The strongest defensible scanner is a reversible hybrid pipeline, not one global enhancement or edge filter. Use restrained point/histogram processing, small denoising kernels, optional mild unsharp masking, targeted periodic-band correction (not broad low-pass), geometric page rectification, directional morphology for table rules, Canny/contour geometry for page corners, and Otsu/adaptive binarization selected by validation. Stain and dust detections should remain review overlays until ground-truth metrics show that automatic classification is safe.

The generated measurements help choose candidate parameters, but they do not satisfy the assignment's external validation requirement. Best synthetic text Dice was about 0.831, below the required ≥0.90; stain IoU was low; model-based comparisons and real DIBCO tests remain to be run. The next evidence required is at least 10 genuinely captured old pages plus DIBCO pages/masks, hand-labelled page corners/stains/lines, OCR reference text, and timings on the lab PC at full target resolution.

Practical validation checklist

- Capture at least 10 old pages with variation in yellowing, stain types, pen/pencil, shadow and table rules; do not use only clean textbook scans.

- Download a DIBCO/DIB Platform subset and its matching ground-truth masks; log version, image names and licensing/usage conditions [1,2].
- Annotate text foreground, page corners, table lines, stains and dust independently; test by page, not by randomly splitting tiles from the same page.
- Run each candidate on the same pages and report mean + range/standard deviation for Dice, IoU, CER, precision/recall, corner error and median/p95 runtime.
- Only claim target compliance after the real test set meets Dice ≥ 0.90 and p95 processing time ≤ 3 s/page (≤ 1 s for the adapted phone workload).

References

[1] Document Image Binarization Platform (DIB Platform), dataset and quality/time assessment. https://dib.cin.ufpe.br/ (accessed 29 October 2026).

[2] TC-11 / CVC, "Document Image Binarization Platform," dataset record describing photographed and real-world document collections. https://tc11.cvc.uab.es/datasets/DIB%20Platform_1 (accessed 29 October 2026).

[3] Y.-H. Lin, W.-C. Chen and Y.-Y. Chuang, "BEDSR-Net: A Deep Shadow Removal Network From a Single Document Image," CVPR 2020. https://openaccess.thecvf.com/content_CVPR_2020/html/Lin_BEDSR-Net_A_Deep_Shadow_Removal_Network_From_a_Single_Document_CVPR_2020_paper.html

[4] M. A. Souibgui et al., "DocEnTr: An End-to-End Document Image Enhancement Transformer," ICPR 2022. https://doi.org/10.1109/ICPR56361.2022.9956101 and https://github.com/dali92002/DocEnTR

[5] OpenCV documentation: histogram equalization/CLAHE, filtering, thresholding, Canny, morphology, contours and geometric transforms. https://docs.opencv.org/4.x/

[6] Tesseract OCR documentation. https://tesseract-ocr.github.io/

[7] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018.

Appendix A — Reproducibility

The companion ZIP contains the script, synthetic inputs and masks, generated figures, CSV metric tables, OCR results and setup instructions. It was executed with Python, OpenCV, NumPy, Matplotlib, Pillow, scikit-image and Tesseract where available. The default benchmark is deterministic (fixed random seed). Example:

```
python smart_scanner_cv_assignment.py --output-dir results --num-pages 12
```

For real photos, place them in a directory and run:

```
python smart_scanner_cv_assignment.py --output-dir results_real --num-pages 12 --input-dir path/to/photos
```

The real-input quick check exports CLAHE and adaptive-threshold images and basic image statistics. Accuracy scores must not be generated until matching ground-truth masks or verified OCR transcripts are supplied.